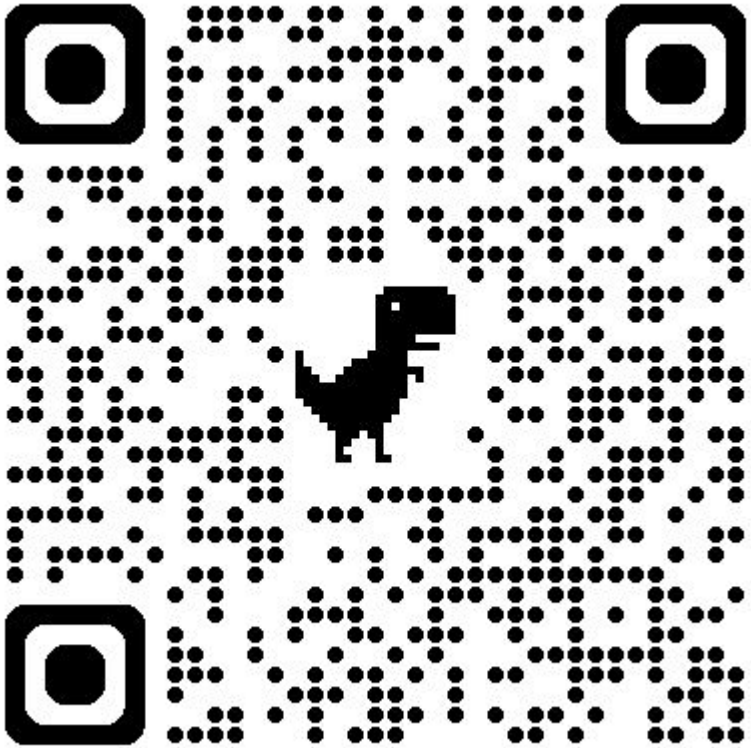


To access all slides for CS010A Labs

Week 3 Lab - CS010A

RGB Values

TA: Sourabha Bharadwaj
Instructors: Prof. Kelly Downey and Saul Gonzalez

Slides borrowed from Alexander Hickerson

Agenda

1. RGB and Gray - What’s the goal?
2. How do you accomplish it?
3. Conditional Statements
4. Logical Operators
5. Real world example
6. RGB problem
7. Questions!

RGB and Gray - What's the goal?

- RGB values range from 0 to 255
- Equal RGB values → shade of gray
  - Examples: (50, 50, 50), (255, 255, 255), (0, 0, 0)
- Gray is the smallest shared value among the 3 colors
  - So, if the 3 colors are (130, 50, 130), then the gray = 50
  - If the 3 are (100, 30, 30), the gray = 30
- To remove gray, subtract the smallest from all 3 numbers
- Let's take a look at how this works



Red

148	162	175	182	189	194	195	193	195	195	197
148	164	174	176	185	189	191	191	196	194	195
144	159	167	176	178	185	188	191	196	194	197
128	147	157	168	173	179	182	184	191	191	192
119	134	148	160	164	170	179	176	181	189	185
145	124	142	151	160	168	169	174	180	182	183
172	120	140	153	157	169	171	178	180	182	182
196	120	129	144	152	158	167	170	177	176	178
204	144	116	134	142	149	155	165	165	170	171

Green

42	43	48	50	53	56	56	53	54	54	54
50	49	51	47	53	55	56	55	59	55	54
51	48	47	49	49	51	50	52	54	51	54
53	48	45	49	50	52	50	48	51	50	50
59	43	43	48	47	48	54	47	49	55	50
100	42	41	42	44	46	45	46	50	52	50
142	47	43	42	39	46	44	48	49	51	49
185	65	44	42	42	43	48	46	50	48	49
209	106	44	42	41	42	44	50	48	50	49

Blue

16	24	32	35	37	40	40	37	37	38	36
19	25	31	28	34	37	38	37	40	35	33
17	23	27	33	32	35	33	36	39	35	37
20	19	23	31	33	34	34	32	36	35	35
29	16	24	33	32	34	39	30	31	38	34
71	11	18	24	30	33	30	30	34	36	34
113	14	16	21	24	32	30	32	33	35	33
156	32	13	20	25	28	33	31	35	33	32
177	72	9	16	22	26	30	35	32	33	32

https://www.slideshare.net/slideshow/03pembentukancitradandigitalisasicitrapptx/260122946

So, How Do We Accomplish This?

Conditional statements, and logical operators

1. Conditional Statements!

- 3 simple statements to remember
  - if, else if, and else
- Common comparisons you can use
  - == (equal to)
  - < / <= (less than, less or equal to)
  - > / >= (greater than, greater or equal to)

2. && Logical Operators

Operator	Symbol in C++	Example	Meaning
AND	&&	(a < b) && (a < c)	True when both sides are true
OR	||	(a < b) || (a < c)	True when at least one side is true
NOT	!	!a	Not true when the operand that follows is true

Let’s Put it Into Practice

- Example:
  - You want to pick a meal plan at the dining halls
  - You have 3 different options and you want to choose the cheapest
  - How do you think you can do it?

1. Initialize variables to hold the values of each meal plan
2. Take in those 3 values
3. Compare them to each other
4. Take the smallest of the 3

RGB Problem

- 1st step: same basic idea as the meal plan problem
- 2nd step: Once you find that smallest value, subtract it from all 3 color values individually

Questions?

Submission Template

Overall sheet:

Code Plan:
1.
2.
3.
…
_____
Test Cases:
1.
2.
3.
…
_____
C++ Code:

C++ Code:

```
#include<iostream>
// other headers
using namespace std;

int main() {
    // initialize variables
    // get user inputs
    // program logic
    // print output
    return 0;
}
```

Test case:

1) Input: 1 2 3
   Expected output: a b c

2) Input: 4 5 6
   Expected output: x y z

3) Input: 0 0 0

   Expected output: Invalid inputs / errors